

Applications Development Project 3 ADPA301

Project Group

I have read, understood and agree to the contents of this document. I accept full responsibility for any errors and omissions in this document.

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Executive Summary

The ElevateED system is a school management platform designed to streamline academic and administrative processes at Mpiyakhe High School. It integrates key school function into a single digital system, improving efficiency, communication and accessibility for Learners, teachers and administrators.

The system supports the full student life cycle, including online enrolment, application approval and class allocation. Daily operations are enhanced through a digital attendance system using a One-Time Pin (OTP) allowing Learners to mark attendance quickly and accurately while enabling attendance tracking and analytics.

Academic support features include access to study material, task and assignments, past paper repository, and a quiz system that allows student to test their knowledge and get immediate results and track their progress.

The system includes extra classes where administrators create additional classes based on grade and subject. Learners can participate in these extra classes and make payments through an integrated payment system. Once payment is confirmed Learners are automatically registered and added to a transport list. A transport management feature allows temporal driver access, route tracking and emergency alerts to ensure Learner safety.

Additional functions include issue reporting and feedback, timetable management, results capturing and automatic report generation with performance analytics

Overall, the ElevateED system creates a centralised and efficient school environment that enhances academic support, improves operational processes and supports better decision-making through data driven insights.

No	Business process	Use case	Sem / Inc	Brief Use Case description	Full Name	Completed Yes / No
1.	<i>Attendance Session Initiation Process</i>	<i>Start Attendance Session</i>	<i>1/2</i>	<i>Teacher initiates attendance using a one-time PIN</i>	<i>Anele</i>	<i>Yes</i>
2.	<i>Student Attendance Capture Process</i>	<i>Submit Attendance via PIN</i>	<i>1/2</i>	<i>Learner enters PIN to mark themselves present</i>	<i>Anele</i>	<i>Yes</i>
3.	<i>Attendance Adjustment and Correction Process</i>	<i>Edit Attendance</i>	<i>1/2</i>	<i>Teacher reviews or corrects attendance in case of any mistakes</i>	<i>Kamva</i>	<i>Yes</i>
4.	<i>Class Attendance Monitoring and Analysis</i>	<i>Review attendance analytics</i>	<i>1/2</i>	<i>Monitor attendance</i>	<i>Kamva</i>	<i>Yes</i>

5	<i>Quiz Content Upload and Management</i>	<i>Upload Quiz and Questions</i>	<i>1/2</i>	<i>Teacher uploads quizzes and questions for Learners</i>	<i>Nosipho</i>	<i>Yes</i>
6	<i>Practice Quiz Management Process</i>	<i>Complete Practice Quiz</i>	<i>1/2</i>	<i>Learner selects a subject and the system generates a practice quiz consisting of multiple-choice and short answer questions</i>	<i>Njabulo</i>	<i>Yes</i>
7.	<i>Extra Class Management</i>	<i>Create Extra Class</i>	<i>1/2</i>	<i>Admin creates extra class for a specific grade and subject</i>	<i>Kaylan</i>	<i>Yes</i>
8.	<i>Extra Class Enrolment</i>	<i>Join Extra Class</i>	<i>1/2</i>	<i>Learner receives info and signs up for the extra class</i>	<i>Kaylan</i>	<i>Yes</i>
9	<i>Payment Management Process</i>	<i>Process Payment for extra class</i>	<i>1/2</i>	<i>Leaner completes payment for an online class</i>	<i>Lwanga</i>	<i>Yes</i>
10.	<i>Transport allocation Process</i>	<i>Generate Transport list</i>	<i>1/2</i>	<i>System generates a list of students eligible for transport for extra classes</i>	<i>Lusanda</i>	<i>Yes</i>
11.	<i>Driver Access Creation Process</i>	<i>Generate Temporary Driver Login</i>	<i>1/2</i>	<i>Admin creates temporary login credentials for a driver</i>	<i>Samkelo</i>	<i>Yes</i>

12.	<i>Driver Access Management Process</i>	<i>Manage Driver Access</i>	<i>1/2</i>	<i>Admin views and controls active temporary driver accounts</i>	<i>Samkelo</i>	<i>Yes</i>
13	<i>Learner pick up management</i>	<i>Pick up Leaners</i>	<i>1/2</i>	<i>The driver collects learners from their designated pick-up points before starting the route</i>	<i>Lwanga</i>	<i>Yes</i>
14	<i>Learner Drop-off Management</i>	<i>Drop off Learners</i>	<i>1/2</i>	<i>The driver drops off learners at their designated destinations after completing the route</i>	<i>Lusanda</i>	<i>Yes</i>
15.	<i>Transport Route Initiation Process</i>	<i>Start Transport Route</i>	<i>1/2</i>	<i>Driver begins transport route for extra classes</i>	<i>Minenhle</i>	<i>Yes</i>
16.	<i>Transport Route Completion Process</i>	<i>End Transport Route</i>	<i>1/2</i>	<i>Driver completes Transport route after class</i>	<i>Minenhle</i>	<i>Yes</i>
17.	<i>Emergency Alert Process</i>	<i>Trigger Emergency Alert</i>	<i>1/2</i>	<i>Driver sends emergency alert with location to admin</i>	<i>Vamile</i>	<i>Yes</i>

Use case number	01
Use case name	<i>Start Attendance Session (OTP)</i>
Brief description	<i>Class teacher initiates attendance using an OTP.</i>
Triggering event	<i>Teacher clicks “Start Attendance”</i>
Actor(s)	<i>Teacher</i>
Preconditions	<i>Teacher is assigned to a class</i>
Flow of activities	1. <i>Teacher navigates to attendance page</i> 2. <i>Teacher click clicks “Start Attendance”</i> 3. <i>OTP is generated</i> 4. <i>Attendance session is created (linked to class & date)</i> 5. <i>Teacher announces OTP</i>

Use case number	02
Use case name	<i>Submit Attendance via OTP</i>
Brief description	<i>Learner enters OTP to mark themselves present.</i>
Triggering event	<i>Learner enters OTP and submits</i>
Actor(s)	<i>Learner</i>
Preconditions	<i>Active attendance session exists for Learners class</i>
Flow of activities	1. <i>Learner opens “Mark Attendance” page</i> 2. <i>Learner enters OTP</i> 3. <i>System validates OTP</i> 4. <i>Learner is marked as present</i> 5. <i>System stores record</i>

Use case number	03
Use case name	<i>Edit Attendance (Manual Override)</i>
Brief description	<i>Teacher reviews or corrects attendance in case of any mistakes.</i>
Triggering event	<i>Error occurrences</i>
Actor(s)	<i>Teacher</i>
Preconditions	<i>Attendance session is completed</i>
Flow of activities	1. <i>Teacher views attendance table for class</i> 2. <i>System displays all Learners</i> 3. <i>Teacher manually edits present/absent where necessary</i> 4. <i>Teacher submits changes</i>

Use case number	04
Use case name	<i>Review Attendance Analytics</i>
Brief description	<i>Monitor attendance</i>
Triggering event	<i>Opens analytics dashboard</i>
Actor(s)	<i>Teacher, Admin</i>
Preconditions	<i>Attendance data exists</i>
Flow of activities	1. <i>Navigates to analytics page</i> 2. <i>System retrieves attendance data for all classes</i> 3. <i>Select filters to (daily, weekly, monthly)</i> 4. <i>Review insights</i>
Use case number	05
Use case name	<i>Upload Quiz and Questions</i>
Brief description	<i>Teacher uploads quizzes and questions for Learners</i>
Triggering event	<i>The teacher decides to upload a new quiz and its questions into the system</i>
Actor(s)	<i>Teacher</i>
Preconditions	<i>Teacher is logged in</i>
Flow of activities	1. <i>Teacher uploads completed spreadsheet</i> 2. <i>System validates file format and content</i> 3. <i>System extracts quiz questions from the file</i> 4. <i>System stores quiz and questions in the database</i> 5. <i>System confirms successful upload</i>

Use case number	06
Use case name	<i>Complete Quiz</i>
Brief description	<i>Learner selects a subject and the system generates a practice quiz consisting of multiple-choice and short answer questions</i>
Triggering event	<i>Learner selects “Quiz” from dashboard</i>
Actor(s)	<i>Learner</i>
Preconditions	<i>Learner is logged into system</i>
Flow of activities	1. <i>Learner selects subject do</i> 2. <i>Starts quiz</i> 3. <i>Completes quiz</i> 4. <i>Submit Quiz</i> 5. <i>Views marks</i> 6. <i>View quiz history</i>

Use case number	07
Use case name	<i>Create Extra Classes</i>
Brief description	<i>Admin creates extra class for a specific grade and subject</i>
Triggering event	<i>Admin selects “Create Class”</i>
Actor(s)	<i>Admin</i>
Preconditions	<i>Admin is logged into system</i>
Flow of activities	1. <i>Enters class details (grade, subject, date, time, location, fee)</i> 2. <i>Saves the extra class</i>

Use case number	08
Use case name	<i>Join Extra Class</i>
Brief description	<i>Learner receives info and signs up for the extra class</i>
Triggering event	<i>Student opens extra class section</i>
Actor(s)	<i>Learner</i>
Preconditions	<i>Learner is logged in</i>
Flow of activities	1. <i>Learner navigates to extra class</i> 2. <i>Views extra class details</i> 3. <i>Clicks join class</i> 4. <i>Makes the payment</i>

Use case number	09
Use case name	<i>Process Payment for an extra class using online payment system</i>
Brief description	<i>Learner makes a Payment for an extra class using online payment system</i>
Triggering event	<i>Learner clicks “Pay Now” after joining extra class</i>
Actor(s)	<i>Learner, system</i>
Preconditions	<i>Learner is logged in</i>
Flow of activities	1. <i>Learner selects “Pay Now” for the chosen class</i> 2. <i>System redirects learner to online payment gateway</i> 3. <i>Learner enters payment details and confirms transaction</i> 4. <i>Payment gateway process payment</i> 5. <i>System receives payment confirmation form payment gateway</i> 6. <i>System confirms learner’s enrolment in the extra class</i> 7. <i>System adds learner to transport list for class</i> 8. <i>System display success message to learner</i>

Use case number	10
Use case name	<i>Generate Transport List</i>
Brief description	<i>System generates a list of students eligible for transport for extra classes</i>
Triggering event	<i>Payment is successfully completed</i>
Actor(s)	<i>System</i>
Preconditions	<i>Payment status of student is updated to “Paid”</i>
Flow of activities	1. <i>System verifies payment status once student has paid.</i> 2. <i>Adds student transport list for the class</i> 3. <i>System updates transport records.</i> 4. <i>Admin and driver can view transport list</i>

Use case number	11
Use case name	<i>Generate Temporary Driver Login</i>
Brief description	<i>Admin creates temporary login credentials for a driver</i>
Triggering event	<i>Create Driver Access</i>
Actor(s)	<i>Admin</i>
Preconditions	<i>Admin is logged in</i>
Flow of activities	1. <i>Admin enters driver details</i> 2. <i>Sets access duration (start and end time)</i> 3. <i>System generates temporary username and password</i> 4. <i>System stores credentials with expiry</i> 5. <i>Admin provides credentials to driver</i>

Use case number	<i>12</i>
Use case name	<i>Manage Driver Access</i>
Brief description	<i>Admin views and controls active temporary driver accounts</i>
Triggering event	<i>Admin opens Driver Management section</i>
Actor(s)	<i>Admin</i>
Preconditions	<i>Temporary drive account exists</i>
Flow of activities	<i>Admin views the list of active driver accounts</i> <i>System displays account details including expiry date</i> <i>Admin selects a specific driver account</i> <i>Admin chooses to deactivate the account</i> <i>System immediately revokes access and updates record</i>

Use case number	<i>13</i>
Use case name	<i>Pick up Learners</i>
Brief description	<i>The driver collects learners from their designated pick-up locations before starting the route</i>
Triggering event	<i>Driver views assigned learners and proceeds to pick them up</i>
Actor(s)	<i>Driver</i>
Preconditions	<i>Diver is logged in</i>
Flow of activities	<i>Driver views assigned learner list</i> <i>Driver confirms learner pick-up</i> <i>System updates learner status to “Picked Up”</i> <i>System records activity</i>

Use case number	<i>14</i>
Use case name	<i>Drop off Learners</i>
Brief description	<i>Driver drops off learners at their designated destinations after completing the route</i>
Triggering event	<i>Transport route ends and driver begins learner drop-off process</i>
Actor(s)	<i>Driver</i>
Preconditions	<i>Transport route is active or completed</i>
Flow of activities	<i>Driver confirms learner drop-off</i> <i>System updates learner status to “Dropped Off”</i> <i>System records drop-off activity</i> <i>System marks route as “Completed”</i>

Use case number	15
Use case name	<i>Start Transport Route</i>
Brief description	<i>Driver begins transport route for extra classes</i>
Triggering event	<i>Driver clicks “Start Route”</i>
Actor(s)	<i>Driver</i>
Preconditions	<i>Driver is logged in and assigned to a class</i>
Flow of activities	1. <i>Driver logs into system</i> 2. <i>View assigned transport list</i> 3. <i>System begins tracking location</i> 4. <i>System updates status route to Active</i>

Use case number	16
Use case name	<i>End Transport Route</i>
Brief description	<i>Driver completes Transport route after class</i>
Triggering event	<i>Driver clicks “End Route”</i>
Actor(s)	<i>Driver</i>
Preconditions	<i>Route is active</i>
Flow of activities	1. <i>Driver clicks “End Route”</i> 2. <i>System stops tracking location</i> 3. <i>System update route status to completed</i> 4. <i>System stores trip history</i>

Use case number	17
Use case name	<i>Trigger Emergency Alert</i>
Brief description	<i>Driver sends emergency alert with location to admin</i>
Triggering event	<i>Driver clicks “Emergency button”</i>
Actor(s)	<i>Driver</i>
Preconditions	<i>Driver is logged in and route is active</i>
Flow of activities	1. <i>Driver clicks the emergency button</i> 2. <i>System captures the driver’s current location</i> 3. <i>System sends an alert to the admin dashboard</i> 4. <i>Admin receives notification with location and message</i> 5. <i>Admin takes necessary action to respond to the emergency</i>